

Kinetic Theory of Gases

5. Real Gases

Andrew's Experiment on Carbon Dioxide

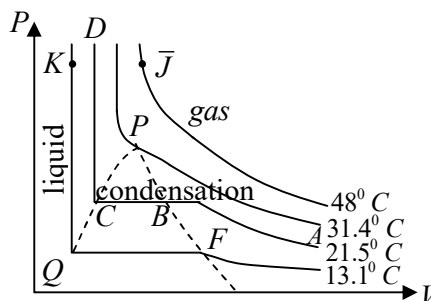
Andrew's experiment investigated the behaviour of CO_2 and analyse the Pressure (P) versus volume (V) at different temperature T .

The observations are following:

1. Above a temperature of about ($T = 48^\circ C$) the CO_2 resemble that of Ideal gas.
2. As temperature lowered, the isotherms exhibit distortion which gradually increases, which is indication from the ideal gas character.
3. At $31.4^\circ C$ a Kink is observed which suggests the gas can be liquified under compression.

4. As temperature is lower further the kink spread into a horizontal line, i.e. compression produces liquification.

From A to B , CO_2 behave as a gas. At the point B the liquification of the gas just starts. The gas condenses at constant pressure from B to C so that liquid and vapour coexist. At C , the gas is completely in the liquid phase. From C to D the slop is very steep since a liquid is almost incompressible.



Conclusion: The temperature at which it becomes possible to liquefy a gas under compression is known as critical temperature (T_c) [In Andrew experiment (T_c) = $48^\circ C$], corresponding pressure and volume is known as critical pressure (P_c) and critical volume (V_c).

A gas can be liquified only if it cooled upto or below its characteristic critical temperature.

There exist a continuity of liquid and gaseous states, i.e. they are two distinct stages at a continuous physical phenomenon.

van der Waals equation of state.

The van der Waals equation for real gases are given by

$$\left(P + \frac{a}{V^2}\right)(V - b) = RT \quad \text{for 1 mole of gas and}$$

$$\left(P + \frac{n^2 a}{V^2}\right)(V - nb) = nRT \quad \text{for } n \text{ mole of gas.}$$

Assumption for real gas:

1. Gas molecules have finite size
2. There are weak interaction force, which depends only upon distance between them.
3. The molecular density is small and the number of collisions with the walls of the container are exactly the same point and finite size molecular.

Example: Under what pressure will carbon dioxide of molar mass M have the density ρ at the temperature T . If given gas is obeying for a Van der Waals gas.

Solution: Assume M is molar mass of the carbon dioxide and V is the volume so $\rho = \frac{M}{V}$

Van der wall equation (for one mole gas):

$$\left(P + \frac{a}{V^2}\right)(V - b) = RT$$

$$\left(P + \frac{\rho^2 a}{M^2}\right)\left(\frac{M}{\rho} - b\right) = RT \Rightarrow P = \frac{RT}{\left(\frac{M}{\rho} - b\right)} - \frac{\rho^2 a}{M^2} \Rightarrow P = \frac{\rho RT}{M - \rho b} - \frac{\rho^2 a}{M^2}$$

Critical Point

The van der Waals equation of state for a gas is given by

$$\left(P + \frac{a}{V^2}\right)(V - b) = RT$$

where P , V and T represent the pressure, volume and temperature respectively, and a and b are constant parameters. At the critical point, where all the roots of the above cubic equation are degenerate means all roots are equal.

In another way mathematically For the critical isotherm is the point of inflection point

On basis of above definition one can find the critical volume V_c , critical pressure P_c and critical temperature T_c for van der waal gas.

For Van der Waals equation

$$\left(P + \frac{a}{V^2}\right)(V - b) = RT$$

$$P = \frac{RT}{V - b} - \frac{a}{V^2} \quad (i)$$

$$\left(\frac{\partial P}{\partial V}\right)_T = 0 \text{ for extremum point}$$

$$\left(\frac{\partial P}{\partial V}\right)_T = -\frac{RT}{(V - b)^2} + \frac{2a}{V^3} = 0 \text{ at } V = V_c, T = T_c \quad (ii)$$

$$\left(\frac{\partial^2 P}{\partial V^2}\right)_T = 0 \text{ for inflection point}$$

$$\frac{2RT}{(V-b)^3} - \frac{6a}{V^3} = 0 \quad \text{at } V = V_c, T = T_c \quad (\text{iii})$$

solving (ii) and (iii)

$$V_c = 3b \quad T_c = \frac{8a}{27Rb}$$

Put value of V_c and T_c one can get $P_c = \frac{a}{27b^2}$

$\frac{RT_c}{P_c V_c} = \frac{8}{3} = c_c$ which is popular known as **critical coefficient** for van der Waals gas.

van der Waals Equation of State and Virial Coefficient

According to virial theorem the equation of state is given by

$$pV = \alpha + \frac{\beta}{V} + \frac{\gamma}{V^2} + \dots \quad (\text{i})$$

Where α, β and γ are first second and third virial coefficient .

For the Ideal gas $\alpha = RT$ and other coefficient are zero.

Virial coefficient for Van der Waals gas

To put van der Waals equation in virial form we first rewrite it as

$$pV = RT \left(1 - \frac{b}{V} \right)^{-1} - \frac{a}{V}$$

Using binomial theorem, we have, $\left(1 - \frac{b}{V} \right)^{-1} = 1 + \frac{b}{V} + \frac{b^2}{V^2} + \dots$

$$\text{Hence, } pV = RT + \frac{RTb - a}{V} + \frac{RTb^2}{V^2} + \dots \quad (\text{ii})$$

As will be noted, van der Waals equation has only three virial coefficients and a comparison with equation (i) yields

$$\alpha = RT, \beta = RTb - a \text{ and } \gamma = RTb^2$$

At the Boyle temperature, the second virial coefficient is zero. Hence, $RT_b b - a = 0$

$$\text{or } T_b = \frac{a}{Rb}$$

From the preceding, section we recall that the critical temperature of a gas obeying van der

Waals equation of state is, $T_C = \frac{8a}{27Rb}$

on comparing these expressions, we get $T_B = \frac{27}{8} T_C = 3.375 T_C$

that is, the Boyle temperature, on the basis of Van der Waals equation, is 3.375 times the critical temperature.